

Refresher Course in Calculus, Probability, and Statistics

Day 1: Calculus

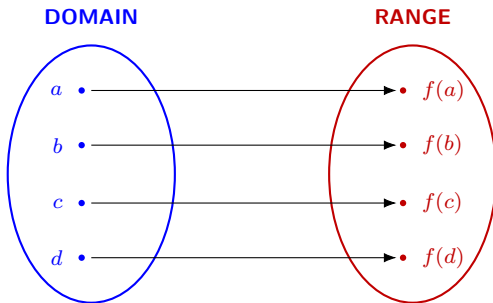
Instructor: Dr. Guillaume Wald

7 September 2026

Function

► Definition:

- A function of a real variable x with domain D is a rule that assigns a unique real number to each number x in D .
- Given a function f , the set of numbers x at which $f(x)$ is defined is called the **domain** of f . The set of values $f(x)$ obtained as x varies in the domain is called the **range** of f .

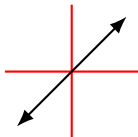


- Example: a supply function $y = a + bp$, where y is output, p is price.

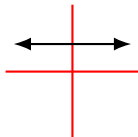
Function Forms

► Examples:

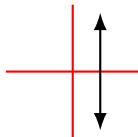
- Linear function: $y = ax + b$
- Quadratic function: $y = ax^2 + bx + c$
- Natural exponential function: $y = e^x$
- Natural logarithmic function: $y = \ln x$



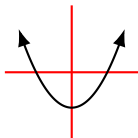
Linear



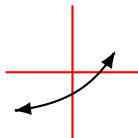
Constant



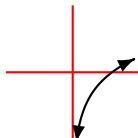
Non function



Quadratic



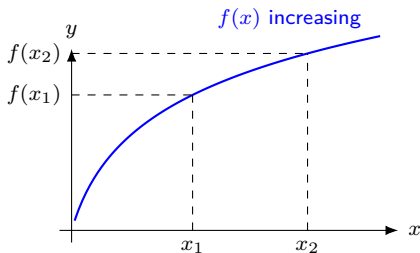
Exponential



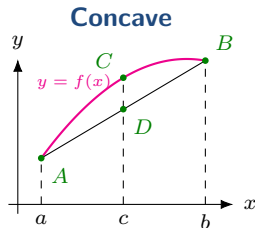
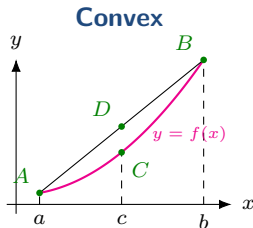
Logarithmic

Geometric Properties of a Function

► Increasing or decreasing



► Concave or convex



Introduction

- ▶ **Differential calculus** is the study of the rate at which quantities change.
 - The primary objects of study are the *derivative* and the *primitive function*.
 - The process of finding a derivative is called *differentiation*.
- ▶ **Integral calculus** can be regarded as the “inverse” of differentiation.
 - The process of reconstructing a function from its derivative is called *integration*.
 - Two kinds of integrals:
 - indefinite integral $\rightarrow$ a function
 - definite integral $\rightarrow$ a number

$\Rightarrow$ Focus on differential calculus and know how to get a derivative.

▶ References:

- Paul's Online Math Notes: <http://tutorial.math.lamar.edu/>

Difference Quotient

► A change in a variable x is denoted: $\Delta x = x_1 - x_0$

► Corresponding change of the function $y = f(x)$:

$$\Delta y = f(x_1) - f(x_0) = f(x_0 + \Delta x) - f(x_0)$$

► **Difference quotient:** change in y per unit of change in x :

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

→ Expresses the *average* change in y for $\Delta x = 1$ on $[x_0, x_1]$

Derivative of a function: Formal Definitions

For a function $y = f(x)$

► At point x_0 :

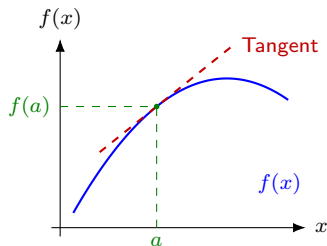
$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

- $f'(x_0)$ is the *instantaneous / marginal* rate of change
- $f' \geq 0$ (> 0) $\iff f$ increasing (strictly increasing)
- $f' \leq 0$ (< 0) $\iff f$ decreasing (strictly decreasing)

► At any point x :

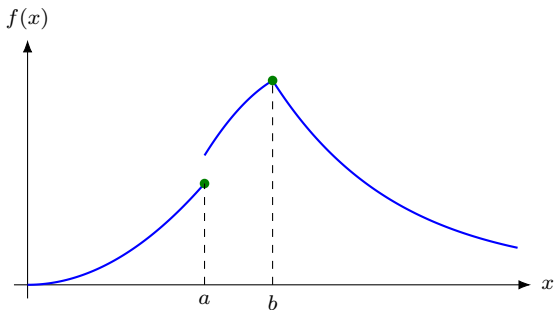
$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$f'(x)$ is a function describing the *instantaneous* rate of change of $y = f(x) \forall x$.



Differentiability and Continuity

- ▶ A function is said to be **differentiable** at a point if its derivative exists at that point.
- ▶ To be differentiable at a point, a function must:
 - be continuous **AND**
 - have a unique tangent



Notation of the Derivative

- The derivative of $y = f(x)$ may be written in several *equivalent* ways:

$$f'(x) \quad y' \quad f_x(x) \quad \frac{dy}{dx} \quad \frac{df}{dx} \quad \frac{df(x)}{dx} \quad \frac{d}{dx}f(x)$$

- Leibniz's notation (with d's)
- preferred in economics
 - emphasises that $f(x)$ is differentiated with respect to (wrt) x
 - can prevent confusion for functions of several variables $f(x_1, \dots, x_n)$ and for higher-order derivatives



Gottfried Wilhelm Leibniz (1646–1716)

Basic Rules for Differentiation

Function	Derivative
$f(x)$	$\frac{df(x)}{dx}$
α	0
αx	α
x^α	$\alpha x^{\alpha-1}$
α^x	$\alpha^x \ln \alpha$
e^x	e^x
$\ln x$	$\frac{1}{x}$

Simple Examples

- Compute $f'(x)$ when $f(x) = x^3$:

- Using the definition (let $\Delta x = h$):

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} \\ &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2 \end{aligned}$$

- Using the power rule $f(x) = x^\alpha \Rightarrow \frac{df(x)}{dx} = \alpha x^{\alpha-1}$: $\frac{dx^3}{dx} = 3x^2$

- Compute $f'(x)$ when $f(x) = \frac{1}{x}$ (note: $\frac{1}{x} = x^{-1}$)

$$\frac{df(x)}{dx} = -1 x^{-2} = -\frac{1}{x^2}$$

Differentiation of Sums, Products, and Quotients

Function	Derivative
$h(x)$	$\frac{dh(x)}{dx}$
$f(x) \pm g(x)$	$f'(x) \pm g'(x)$
$f(x) \cdot g(x)$	$f'(x) \cdot g(x) + f(x) \cdot g'(x)$
$\frac{f(x)}{g(x)}$	$\frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$

Chain Rule for Differentiation

- ▶ **Composite function** (or function of a function): a function in which y is a function of u and u in turn is a function of x .
- ▶ Derivative of y wrt x :

$$y = f(u) \quad \text{and} \quad u = g(x) \quad \implies \quad y' = f'(u) \cdot g'(x) = f'[g(x)] \cdot g'(x)$$

$$\text{or:} \quad \frac{dy}{dx} = \frac{df}{du} \cdot \frac{dg}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

- ▶ Example: $f(x) = (1 - x^3)^5$, with $g(x) = 1 - x^3$ and $f(u) = u^5$:

$$f'(x) = 5(1 - x^3)^4 \cdot (-3x^2) = -15x^2(1 - x^3)^4$$

Higher-Order Derivatives

- ▶ $f'(x)$: **first derivative** of f
- ▶ $f''(x)$: **second-order derivative** of f

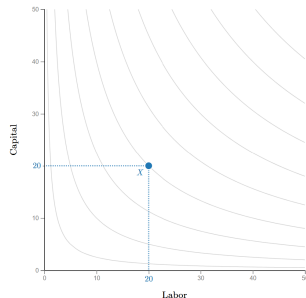
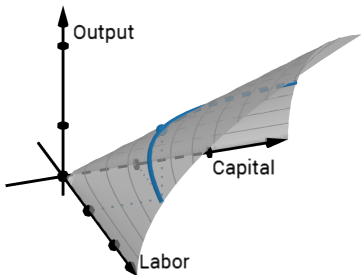
$$f''(x) = \frac{d}{dx} \left(\frac{df(x)}{dx} \right) = \frac{d^2 f(x)}{dx^2}$$

- $f''(x) > 0 \iff f$ convex (at x)
- $f''(x) < 0 \iff f$ concave (at x)

- ▶ $f'''(x) = \frac{d^3 f(x)}{dx^3}$: **third-order derivative** of f
- ▶ $f''''(x) = \frac{d^4 f(x)}{dx^4}$: **fourth-order derivative** of f
- ▶ ...

Multi-variable Functions

- ▶ With functions of several variables, one can measure the rate of change of y wrt x_1 while x_2 is held constant.
- ▶ NB: In economics, “*held constant*” is used equivalently to “*all other things being equal*” or “*ceteris paribus*”.



Cobb-Douglas production function with parameters $A = 1$, $a = 0.5$, $K = 0.5$

Set your own parameters at:

https://www.econgraphs.org/graphs/firm/technology/cobb_douglas_production_3d

Multivariable Functions and Partial Derivatives

- ▶ For functions of several variables $y = f(x_1, x_2, \dots)$, the (first-order) **partial derivative**:
 - is defined as $\frac{\partial y}{\partial x_1}$: (first-order) partial derivative of $f(x_1, x_2, \dots)$ with respect to x_1
 - ∂ (read: “(partial) dee” or “partial”) has the same function as d in *Leibniz notation*
 - measures the rate of change of y wrt x_1 while x_2 is held constant (“*ceteris paribus*”)
- ▶ Notations: $\frac{\partial y}{\partial x_1}$, $\frac{\partial f}{\partial x_1}$, $f_1(x_1, x_2, \dots)$, $f_{x_1}(x_1, x_2, \dots)$, y_1
- ▶ Partial differentiation wrt one of the independent variables follows the **same rules** as ordinary differentiation, with the other variables **treated as constants**.

Second-Order Partial Derivatives

- ▶ Second-order **direct** partial derivatives of $y = f(x_1, x_2)$:

$$\text{wrt } x_1 : f_{11} = f_{x_1 x_1} = \frac{\partial}{\partial x_1} \left(\frac{\partial y}{\partial x_1} \right) = \frac{\partial^2 y}{\partial x_1^2}$$

$$\text{wrt } x_2 : f_{22} = f_{x_2 x_2} = \frac{\partial}{\partial x_2} \left(\frac{\partial y}{\partial x_2} \right) = \frac{\partial^2 y}{\partial x_2^2}$$

- ▶ **Cross (or mixed)** partial derivatives:

$$f_{12} = f_{x_1 x_2} = \frac{\partial}{\partial x_1} \left(\frac{\partial y}{\partial x_2} \right) = \frac{\partial^2 y}{\partial x_1 \partial x_2}$$

$$f_{21} = f_{x_2 x_1} = \frac{\partial}{\partial x_2} \left(\frac{\partial y}{\partial x_1} \right) = \frac{\partial^2 y}{\partial x_2 \partial x_1}$$

- If both cross partial derivatives are continuous, they are identical: $f_{12} = f_{21}$
→ *Young's theorem*.

Exponential Functions

- ▶ Exponential function:

$$f(x) = a^x, \quad a > 0 \quad (a : \text{base}, x : \text{exponent})$$

- ▶ **Properties:**

- Same base:

$$a^x \cdot a^y = a^{x+y}$$

$$\frac{a^x}{a^y} = a^{x-y}$$

$$(a^x)^y = a^{x \cdot y}$$

- Same exponent:

$$a^x \cdot b^x = (ab)^x$$

$$\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$$

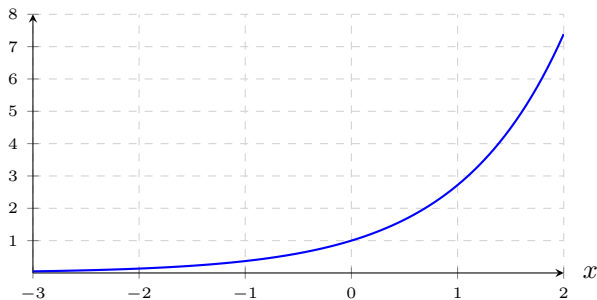
- ▶ **Derivative:** $f'(x) = a^x \ln a$

Natural Exponential Function

- **Natural exponential** function:

$$f(x) = e^x = \exp(x), \quad \text{with } e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828\dots, \text{ known as "Euler's number"}$$

- **Derivatives:** $f'(x) = e^x$, $f''(x) = e^x$, ...



Logarithmic Functions

- ▶ Logarithmic function with base a :

$$f(x) = \log_a x, \quad a > 0, a \neq 1 \quad (a : \text{base}, \log_a x : \text{exponent})$$

Definition: $\log_a x$ is the exponent to which a must be raised to obtain x

$$x = a^y \iff y = \log_a x$$

- ▶ Properties:

- $a^{\log_a x} = \log_a a^x = x$

- $\log_a(x \cdot y) = \log_a x + \log_a y$

- $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$

- $\log_a x^n = n \cdot \log_a x$

- $\iff \log_a \sqrt[n]{x} = \log_a x^{\frac{1}{n}} = \frac{1}{n} \cdot \log_a x$

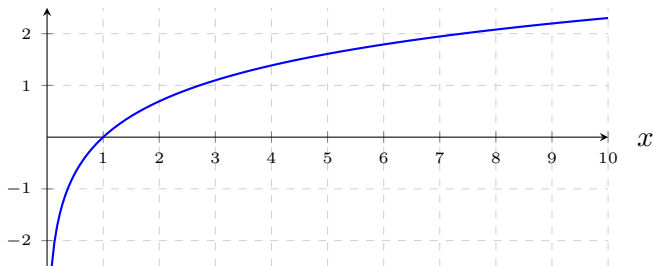
- $\log_a x = (\log_a c)(\log_c x)$

- $\iff \log_c x = \frac{\log_a x}{\log_a c}$

- ▶ Derivative: $f'(x) = \frac{1}{x \ln a}$

Natural Logarithmic Function

- ▶ Natural logarithmic function: $f(x) = \log_e x = \ln x$
- ▶ Inverse function of e^x : $e^{\ln x} = \ln(e^x) = x$
- ▶ Derivatives: $f'(x) = \frac{1}{x}$, $f''(x) = -\frac{1}{x^2}$, $f'''(x) = \frac{2}{x^3}$, $\dots$



Logarithmic Differentiation

- Logarithmic differentiation: $\frac{d \ln f(x)}{dx} = \frac{f'(x)}{f(x)}$ iff $f(x)$ is differentiable

Demonstration

- Suppose that $y = \ln f(x)$ and $f(x)$ is differentiable
- This is a composite function $y = g[f(x)]$, such that $g(u) = \ln u$ and $u = f(x)$
- Differentiate both functions wrt u and x respectively:
 $g' = \frac{1}{u}$ and $u' = f'(x)$
- Apply the chain rule $y' = g'(u) \cdot f'(x) \Rightarrow y' = g' \cdot u' = \frac{1}{u} f'(x) = \frac{f'(x)}{u}$.
- Recall that $u = f(x)$ and substitute:

$$y' = \frac{f'(x)}{u} = \frac{f'(x)}{f(x)}$$

- Use this property to differentiate awkward functions:

$$f'(x) = \frac{df(x)}{dx} = \frac{d \ln f(x)}{dx} f(x)$$

Logarithmic Differentiation (continued)

Example 1: find $f'(x)$

► $f(x) = x^x$

- Taking the logarithm:

$$\ln f(x) = \ln x^x = x \ln x$$

- Differentiate both sides wrt x (using logarithmic differentiation on the lhs):

$$\frac{f'(x)}{f(x)} = 1 \cdot \ln x + x \left(\frac{1}{x} \right) = \ln x + 1$$

- Multiply both sides by $f(x)$:

$$f'(x) = (\ln x + 1) x^x$$

Logarithmic Differentiation (continued)

Example 2: find $f'(x)$

► $f(x) = A(x)^\alpha B(x)^\beta$

- Taking the logarithm:

$$\ln f(x) = \ln(A(x)^\alpha B(x)^\beta) = \alpha \ln A(x) + \beta \ln B(x)$$

- Differentiate both sides wrt x (using logarithmic differentiation):

$$\frac{f'(x)}{f(x)} = \alpha \frac{1}{A(x)} A'(x) + \beta \frac{1}{B(x)} B'(x) = \alpha \frac{A'(x)}{A(x)} + \beta \frac{B'(x)}{B(x)}$$

- Multiply both sides by $f(x)$:

$$f'(x) = \left(\alpha \frac{A'(x)}{A(x)} + \beta \frac{B'(x)}{B(x)} \right) A(x)^\alpha B(x)^\beta$$

Elasticities

- ▶ Elasticity of $y = f(x)$ with respect to x
 - ratio between the relative change in y and the relative change in x
 - dimensionless measure of association

$$\varepsilon_{yx} = \frac{\frac{dy}{y}}{\frac{dx}{x}} = \frac{dy}{dx} \cdot \frac{x}{y}$$

- $f(x)$ is *elastic* at a point when $\varepsilon_{yx} > 1$
of unit elasticity at a point when $\varepsilon_{yx} = 1$
inelastic at a point when $\varepsilon_{yx} < 1$

- ▶ Generally: $\varepsilon_{yx} = \frac{dy}{dx} \cdot \frac{x}{y} = \frac{d \ln y}{d \ln x}$ because $d \ln y = \frac{dy}{y}$ and $d \ln x = \frac{dx}{x}$

- ▶ Partial elasticities for functions of several variables $y = f(x_1, \dots, x_n)$ are defined similarly:

$$\varepsilon_{yx_1} = \frac{\partial y}{\partial x_1} \cdot \frac{x_1}{y}, \quad \dots, \quad \varepsilon_{yx_n} = \frac{\partial y}{\partial x_n} \cdot \frac{x_n}{y}$$

Elasticities (continued)

Example: find the elasticity of Y with respect to K for $Y = AK^\alpha L^\beta$

► Definition of elasticity: $\varepsilon_{YK} = \frac{\partial Y}{\partial K} \cdot \frac{K}{Y}$

- $\frac{\partial Y}{\partial K}$ is the partial derivative of Y wrt K : $\frac{\partial Y}{\partial K} = \alpha AK^{\alpha-1} L^\beta$
- $\frac{K}{Y}$ is simply the ratio of K and Y : $\frac{K}{Y} = \frac{K}{AK^\alpha L^\beta} = \frac{1}{AK^{\alpha-1} L^\beta}$
- Combining the two terms:

$$\varepsilon_{YK} = \frac{\partial Y}{\partial K} \cdot \frac{K}{Y} = \alpha AK^{\alpha-1} L^\beta \cdot \frac{1}{AK^{\alpha-1} L^\beta} = \alpha.$$

► Alternative: log transformation

- $\ln Y = \ln(AK^\alpha L^\beta) = \ln A + \alpha \ln K + \beta \ln L$
- Taking the partial derivative of $\ln Y$ wrt $\ln K$: $\frac{\partial \ln Y}{\partial \ln K} = \alpha.$

Derivatives (further information and exercises)

- ▶ More comprehensive discussion of differentiation (on Paul Dawkins' website)
 - One-variable derivatives:
 - Explanation:
<https://tutorial.math.lamar.edu/Classes/CalcI/DerivativeIntro.aspx>
 - Exercises:
<https://tutorial.math.lamar.edu/Problems/CalcI/DerivativeIntro.aspx>
 - Partial derivatives:
 - Explanation:
<https://tutorial.math.lamar.edu/Classes/CalcIII/PartialDerivsIntro.aspx>
 - Exercises:
<https://tutorial.math.lamar.edu/Problems/CalcIII/PartialDerivsIntro.aspx>
 - Applications:
 - Explanation:
<https://tutorial.math.lamar.edu/Classes/CalcI/DerivAppsIntro.aspx> &
<https://tutorial.math.lamar.edu/Classes/CalcIII/PartialDerivAppsIntro.aspx>
 - Exercises:
<https://tutorial.math.lamar.edu/Problems/CalcI/DerivAppsIntro.aspx> &
<https://tutorial.math.lamar.edu/Problems/CalcIII/PartialDerivAppsIntro.aspx>

Indefinite Integral

- Suppose we do not know the primitive function H .

- All we know is its derivative: $H'(t) = \frac{dH}{dt} = t^{-\frac{1}{2}}$
- We can use the differentiation rule: $f(x) = x^\alpha \Rightarrow f'(x) = \alpha x^{\alpha-1}$
- And reverse it: $f'(x) = x^\alpha \Rightarrow f(x) = \frac{1}{\alpha+1} x^{\alpha+1}$

- Possible solution for H :

- $H(t) = 2t^{\frac{1}{2}}$...but what about $H(t) = 2t^{\frac{1}{2}} + 99$ or $H(t) = 2t^{\frac{1}{2}} + 27000$?

- Because all constants disappear, the indefinite integral $H(t) = 2t^{\frac{1}{2}} + c$ is a *group* of functions, where c is an arbitrary constant

- To solve for H explicitly we need additional information:
an *initial* or *boundary condition* (the state of H at a given value of t), e.g:

$$H(0) = 33 \Rightarrow 33 = 2 \cdot 0^{\frac{1}{2}} + c \iff c = 33.$$

$$\Rightarrow H(t) = 2t^{\frac{1}{2}} + 33$$

Indefinite Integral

- ▶ Definition of the **indefinite integral**:

$$\int f(x) \, dx = F(x) + c \quad \text{when } F'(x) = f(x)$$

- ▶ Basic vocabulary:

- symbol $\int$: integral sign
- function $f(x)$: integrand
- x : variable of integration (indicated by dx)
- c : constant of integration

- ▶ We say *indefinite* because the technique of integration is not applied to any numbers, borders, or limits

→ the result is usually a *function*.

Common Integrals

Function	Anti-derivative
$\int x^{\alpha} dx$	$\frac{1}{\alpha + 1} x^{\alpha+1} + c \quad \text{if } \alpha \neq -1$
$\int \alpha dx = \int \alpha x^0 dx$	$\alpha x + c$
$\int dx = \int 1 x^0 dx$	$x + c$
$\int \frac{1}{x} dx$	$\ln x + c$
$\int \frac{1}{\alpha x + \beta} dx$	$\frac{1}{\alpha} \ln \alpha x + \beta + c$
$\int e^x dx$	$e^x + c$
$\int e^{\alpha x + b} dx$	$\frac{1}{\alpha} e^{\alpha x + b} + c \quad \text{if } \alpha \neq 0$
$\int \alpha^x dx$	$\frac{1}{\ln \alpha} \alpha^x + c \quad \text{if } \alpha > 0 \text{ and } \alpha \neq 1$

Indefinite Integral: Properties

- ▶ Multiplicative constants

$$\int \alpha f(x) \, dx = \alpha \int f(x) \, dx$$

- ▶ Multiplicative constant -1

$$\int -f(x) \, dx = - \int f(x) \, dx$$

- ▶ Sums and differences

$$\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx$$

Indefinite Integral: Examples

Example 1: evaluate the integral

$$\begin{aligned}\int \left(x^4 - \frac{3}{2x} + e^{2x} - 9 \right) dx &= \int x^4 dx - \frac{3}{2} \int \frac{1}{x} dx + \int e^{2x} dx - \int 9 dx \\ &= \frac{1}{5}x^5 - \frac{3}{2} \ln |x| + \frac{1}{2}e^{2x} - 9x + C\end{aligned}$$

Example 2: a common mistake — spot the two errors

$$\begin{aligned}\int x^4 - \frac{3}{2x} dx + e^{2x} - 9 &= \int x^4 dx - \frac{3}{2} \int \frac{1}{x} dx + e^{2x} - 9 \\ &= \frac{1}{5}x^5 - \frac{3}{2} \ln |x| + e^{2x} - 9 + C\end{aligned}$$

Definite Integral

- ▶ Definite integrals involve integrating a function between two values (the “limits”)
- ▶ Definition of the definite integral:

$$\int_a^b f(x) \, dx = \left[F(x) \right]_a^b = F(b) - F(a)$$

where F is any function satisfying $F'(x) = f(x)$ for all $x \in [a, b]$

- ▶ (Geometric) interpretation
 - if $f(x) \geq 0$ over $[a, b]$, then

$$\int_a^b f(x) \, dx$$

is the area between the graph of f , the x -axis and the lines $x = a$ and $x = b$.

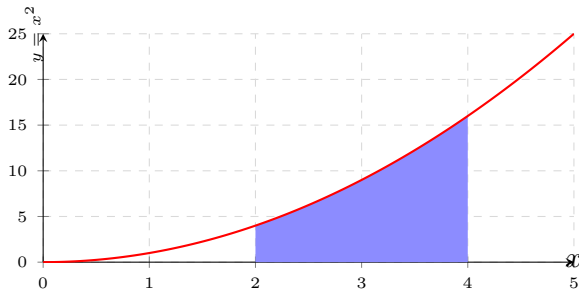
- if $f(x) \leq 0$ over $[a, b]$, then

$$-\int_a^b f(x) \, dx$$

is the area between the graph of f , the x -axis and the lines $x = a$ and $x = b$.

Definite Integral: Geometric Interpretation

Example: find the area between the x -axis and $f(x) = x^2$ over $[2, 4]$ (shaded area in the figure below)



$$\int_2^4 x^2 dx = \left[\frac{1}{3}x^3 \right]_2^4 = \frac{1}{3}4^3 - \frac{1}{3}2^3 = \frac{64}{3} - \frac{8}{3} = \frac{56}{3} \approx 18.66$$

Definite Integral: Properties

$$\blacktriangleright \int_a^b f(x) \, dx = - \int_b^a f(x) \, dx$$

$$\blacktriangleright \int_a^a f(x) \, dx = 0$$

$$\blacktriangleright \int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx, \quad a \leq c \leq b$$

$$\blacktriangleright \int_a^b \alpha f(x) \, dx = \alpha \int_a^b f(x) \, dx$$

$$\blacktriangleright \int_a^b [f(x) + g(x)] \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx$$

Definite Integral: Present Value of a Future Cash Flow

- ▶ Net present value: current value of a future reward or payment

$$A = Ve^{-rt}$$

A : net present value V : nominal value r : discount rate t : time to reception

→ A dollar received later is worth less today, as it could be invested and grow at rate r .

- ▶ What about a flow/stream of future payments?
 - Sum of all discounted payments over all periods

$$\Pi = \int_0^{\tau} Ve^{-rt} dt = V \int_0^{\tau} e^{-rt} dt = V \left[-\frac{1}{r} e^{-rt} \right]_0^{\tau} = \frac{V}{r} (1 - e^{-r\tau})$$

- assume $r = 0.06$, $\tau = 5$ and $V = \$3000$
- present value of the future cash flow: $\Pi = \frac{\$3000}{0.06} (1 - e^{-0.06 \cdot 5}) \approx \$12,959$.

→ Counted naively, five years of payments look like $5 \times \$3000 = \$15,000$. The \$2,041 gap is precisely the **cost of waiting**.

Why a *Natural* Exponential?

- ▶ **Interest = growth proportional to current value.** In continuous time this is a differential equation, with a unique solution:

$$\frac{dA}{dt} = r A \quad \implies \quad A(t) = A_0 e^{rt}.$$

Since e^t is the *only* function equal to its own derivative, once growth is proportional the base e is *forced*, not chosen.

- ▶ **Why e and no other base?** Only for base e does the exponent's r equal the true instantaneous rate:

$$\frac{d}{dt} a^t = a^t \ln a \quad \implies \quad \text{any } a \neq e \text{ smuggles in a stray } \ln a.$$

- ▶ **Where e comes from: continuous compounding.**

$$\left(1 + \frac{r}{n}\right)^{nt} \xrightarrow{n \rightarrow \infty} e^{rt}.$$

Discounting is simply this run *backwards* in time: $A = V e^{-rt}$.

Integrals (further information and exercises)

- ▶ More comprehensive discussion of integration techniques (on Paul Dawkins' website)
 - Basics and integration by substitution:
 - Explanation:
<http://tutorial.math.lamar.edu/Classes/CalcI/IntegralsIntro.aspx>
 - Exercises:
<http://tutorial.math.lamar.edu/Problems/CalcI/IntegralsIntro.aspx>
 - More techniques:
 - Explanation:
<http://tutorial.math.lamar.edu/Classes/CalcII/IntTechIntro.aspx>
 - Exercises:
<http://tutorial.math.lamar.edu/Problems/CalcII/IntTechIntro.aspx>
 - Multiple integrals:
 - Explanation:
<http://tutorial.math.lamar.edu/Classes/CalcIII/MultipleIntegralsIntro.aspx>
 - Exercises:
<http://tutorial.math.lamar.edu/Problems/CalcIII/MultipleIntegralsIntro.aspx>